

04_02: RQ1 Dataset Distribution, Characteristics, and Source Grouping Matrix

1. Objective

This document addresses **RQ1** by summarizing the characteristics, distribution, and source grouping of datasets used in the 110 included studies. It also supports RQ2 by identifying the applicability and limitations of common dataset families, including synthetic benchmarks, real-world/proprietary logs, multi-source datasets, and domain-specific datasets.

The following table provides the quantitative breakdown of the dataset grouping types identified in the literature, maintaining the specific sub-type details for each category.

2. High-Level Dataset Grouping Summary

The 110 included studies were first grouped into four broad dataset categories. This high-level grouping shows the overall distribution of dataset usage across the reviewed literature and highlights the dominance of CERT-based benchmark studies.

Dataset Grouping Type	Total Count	Percentage (%)
Single Source: CERT Benchmark	67	60.91%
Multi-Source Fusion	23	20.91%
Single Source: Real-world / Proprietary	10	9.09%
Single Source: Other / Specialized Synthetic	10	9.09%
Total	110	100%

As shown in Table 1, the CERT benchmark is the most frequently used dataset family, accounting for 60.91% of the reviewed studies. Multi-source fusion datasets represent the second largest group, while real-world/proprietary and other specialized datasets appear less frequently. This distribution shows that the field still relies heavily on public synthetic benchmarks.

3. Detailed Dataset Distribution by Sub-Type

Dataset Grouping Type	Dataset Family / Sub-type	Total Count	Percentage (%)
Single Source	CERT (r4.1–r6.2)	67	60.91%
	Real-world / Proprietary: Enterprise	4	3.64%
	Real-world / Proprietary: Healthcare	2	1.82%
	Real-world / Proprietary: Telemetry/Network	2	1.82%
	Real-world / Proprietary: Military/Org	2	1.82%
	Specialized Synthetic / Org logs	4	3.64%
	Enron email corpus	2	1.82%
	UEBA and Ontology testbeds	2	1.82%
	Special Repository / Biometric	2	1.82%
	CERT + LANL + TWOS	4	3.64%
Multi-Source Fusion	CERT + Network/IDS (NSL-KDD, etc.)	3	2.73%
	Enron + Other (TWOS, CERT, etc.)	3	2.73%
	Biometric + Log/Simulated data	3	2.73%
	CERT + Specialized (Wiki, CCTV, etc.)	4	3.64%
	IoT / Industrial + Network logs	3	2.73%
	Other Multi-source / UEBA / Synthetic	3	2.73%
Total		110	100%

Table 2 shows that single-source CERT studies form the largest sub-category. Real-world/proprietary datasets appear across enterprise, healthcare, telemetry/network, and military or organizational contexts, but their total use remains limited. Multi-source fusion studies combine datasets such as CERT, LANL, TWOS, Enron, NSL-KDD, UNSW-NB15,

biometric logs, and IoT/industrial sources to improve behavioral coverage, although this also increases preprocessing complexity.

4. Characteristics of Common Insider Threat Datasets

Table 3 summarizes the main characteristics of common insider threat datasets discussed in the reviewed studies. It reports dataset scale, log sources or modalities, labeling or threat-scenario information, and the main research use of each dataset. This table supports RQ1 by showing how dataset characteristics differ across benchmark, real-world, simulated, and expert-informed sources.

Dataset	Main Characteristics	Log Sources / Modalities	Labeling / Threat Scenario Information	Main Use
CERT	Simulates 1000–4000 users depending on version; 501 days of raw data	Logon, file, HTTP, email, device, LDAP, psychometric records	Predefined insider scenarios, including data exfiltration, intellectual property theft, and IT sabotage	Benchmark insider threat detection
LANL	More than one billion logs; 12,425 users; 17,684 computers; 58 days	Authentication, network flow, DNS	749 malicious logons from 98 compromised accounts	Authentication anomaly and lateral movement
TWOS	Five days of real-machine activity; one traitor; 12 masqueraders	Login, network, email, keystroke, mouse, display data	Masquerader and traitor behavior	Multimodal behavioral detection
Enron	About 250,000 emails from 150 employees over four years	Email communication and financial information	Requires additional filtering and labeling for insider threat detection	Text and communication analysis
PicoDomain	Controlled network with domain controller, 5 workstations, firewall	Zeek authentication, anomaly, file, and system logs	Used with CERT in some studies	Small-scale network and behavioral evaluation
HCT	175 users; 13,446 shell commands in JSON format	Shell command activity	Five user/threat categories	Command-sequence behavior analysis
SEA	750,000 logs from 50 users, including 29 malicious users	User activity logs	Includes malicious user behavior	Temporal insider behavior analysis
Purdue UNIX	Simulates eight UNIX users over two years	UNIX session and command activity	Session-based behavioral records	Long-term user behavior analysis
UMDWikipedia	About 770,000 edits from 33,576 users; 17,027 blocklisted users	Wikipedia edit activity	Allowlist and blocklist behavior	Abnormal collaborative behavior detection
SOFIT	Survey-based expert-knowledge dataset	Expert judgments on insider threat indicators	Likelihood-based indicator information	Expert-informed insider risk analysis
Real-world/proprietary logs	Enterprise, healthcare, telemetry, commercial, military, or organizational records	Varies by organization and domain	Often requires expert labeling	Operational validation
Simulated synthetic logs	Scenario-specific organizational or healthcare simulations	Simulated organizational or domain-specific logs	Researcher-defined scenarios	Controlled evaluation

The datasets in Table 3 differ substantially in scale, modality, labeling quality, and intended use. CERT provides broad enterprise-like behavioral logs and predefined insider scenarios, while LANL is more suitable for authentication and lateral-movement analysis. Enron supports

communication analysis, whereas SOFIT provides expert-informed risk indicators. Real-world and proprietary logs offer stronger operational realism but are difficult to reproduce because of privacy and access restrictions.

5. Specialized IoT, IDS, and Biometric Datasets

Table 4 summarizes specialized datasets used in IoT, intrusion detection, physical-security, and biometric insider threat contexts. These datasets extend insider threat detection beyond conventional enterprise activity logs.

Dataset	Domain	Key Characteristics	Use in Included Studies
UNSW-NB15	Network intrusion detection	9 attack scenarios, 49 traffic features, supports binary and multi-class classification	Used for threat detection and classification tasks
X-IloTID	Industrial Internet of Things / attack simulation	Designed for cyberattack simulation and testing in IIoT environments	Combined with NS-3 simulated data to model unique attack scenarios
CIC-IoT 2023	Internet of Things network security	33 Internet of Things attack types collected from 105 real Internet of Things devices	Used with UNSW-NB15 for Internet of Things threat detection in cloud environments
MobileNetV2 Dataset	Physical security / CCTV	CCTV video frames used for behavior recognition and motion analysis	Combined with CERT to detect physical security breaches
SapiMouse	Mouse dynamics	Builds behavioral profiles using movement, dragging, press, and release events from 120 users	Used for insider user authentication
Open-source behavioral dataset	Mouse movements and keystrokes	Captures user interaction behavior	Evaluated in Cooja simulator using N-grams and Jaccard coefficient
KeyRecs	Keystroke dynamics	Simulates password typing patterns with predefined lengths	Used to detect malicious authentication events
GREYC-NISLAB	Keystroke dynamics	Designed for keystroke biometric authentication and typing patterns	Used to improve testing of machine learning authentication models
CMU Keystroke Dataset	Keystroke behavior	Contains authentic user typing data for password-entry tasks	Combined with synthetic datasets for broader model validation

6. Study-by-Study Dataset Mapping for the 110 Included Studies

Table 5 maps each included study to its dataset source grouping and dataset name. This table provides the audit trail behind the counts reported in Tables 1 and 2.

#	Author (Year)	Dataset Source Grouping	Dataset Name(s)
1	Adun (2023)	Single Source	CERT (Kaggle v1)
2	Ahmad (2025)	Single Source	Real-world Production Logs (Wazuh/Windows)
3	Ahmadi (2025)	Single Source	Anonymized Enterprise Logs
4	Alabdulkreem (2022)	Single Source	Aggregated Internal Logs (Logon, Email, etc.)
5	Ali (2025)	Single Source	US Army Insider Threat Hub (Narrative Text)
6	Al-Mhiqani (2021)	Single Source	CERT r4.2
7	Al-Mhiqani (2022)	Single Source	CERT r4.2

8	Almusawy (2024)	Single Source	CERT r4.2
9	Al-Shehari and Alsawail (2021) One-Hot Encoding	Single Source	CERT r4.2
10	Al-shehri Abdullah (2022)	Single Source	CERT r4.2 (32M events)
11	Al-Shehari (2023) [IF]	Single Source	CERT r4.2
12	Al-Shehari (2023) [Resampling]	Single Source	CERT r4.2
13	Al-Shehari (2024) [CNN]	Single Source	CERT r4.2
14	Al-Shehari (2024) [LOF]	Single Source	CERT r4.2
15	AlSlaiman (2023)	Single Source	CERT r4.2
16	Amiri-Zarandi (2023)	Single Source	CERT r4.2
17	Amuda (2022)	Single Source	CERT v4.2
18	Anakath (2022)	Multi-Source	Open-source Biometric + Simulated Logs
19	Anju (2024)	Single Source	CERT r4.2 & r6.2
20	Asha (2024)	Single Source	CERT (CMU) v3.2
21	Budžys (2024)	Multi-Source	CMU Keystroke + GREYC- NISLAB
22	Cai (2024)	Single Source	CERT r4.2 & r5.2
23	Ding (2024)	Single Source	CERT v4.2
24	Dong (2025)	Single Source	CERT r5.2 & r6.2
25	Dosh (2021)	Single Source	CERT r4.1 & r4.2
26	Duan (2024)	Single Source	CMU-CERT r4.2
27	Erola (2022)	Single Source	Real-world Enterprise Logs (3 Companies)
28	Fei (2023)	Multi-Source	CERT r4.2 + LANL
29	Feng (2025)	Single Source	CERT r4.2
30	Ferraro (2025)	Multi-Source	PicoDomain + CERT r5.2
31	Gayathri (2024)	Single Source	CERT r4.2 & r5.2
32	Gayathri R (2025)	Single Source	CERT v4.2 & v5.2
33	Gayathri B (2025)	Multi-Source	CERT + NSL-KDD + UNSW- NB15
34	Gonzales (2025)	Multi-Source	Synthetic Action Databases + UniProt
35	Greitzer (2021)	Single Source	SOFIT Ontology Testbed
36	Rahman (2022)	Single Source	CERT r4.2
37	Haq (2022)	Single Source	Enron Email
38	He (2022)	Multi-Source	Enron + monkey.org + CERT r6.2
39	He (2024)	Multi-Source	Enron + monkey.org + CERT r6.2
40	Hong (2023)	Single Source	CERT r4.2
41	Huang (2025)	Multi-Source	KDD-UEBA + NSL-UEBA + CIC- UEBA + KDD-UNBLogs
42	Hurst (2022)	Single Source	UK Hospital EHR Logs
43	Jaiswal (2024)	Single Source	CERT r4.2 & r5.2

44	Janjua (2021)	Multi-Source	Enron + TWOS
45	Jyosthna (2021)	Single Source	CERT r6.1
46	Kabir (2023)	Single Source	CERT r4.2
47	Kamatchi (2025)	Multi-Source	Simulated (NS-3) + X-IloTID
48	Kan (2023)	Single Source	CERT r4.2
49	Kim (2023)	Single Source	CERT (Scenarios 1-3)
50	Kong (2025)	Multi-Source	CERT r4.2 + SEA + HCT + PU
51	Kotb (2025)	Single Source	CERT r4.2 (Full)
52	Lavanya (2024)	Multi-Source	CERT r6.1 & r6.2 and IoT enabled institutional log records
53	Le (2021) [Exploring]	Single Source	CERT r4.1, r4.2, r5.1, r5.2, r6.2
54	Le (2021) [Ensembles]	Multi-Source	CERT + LANL + TWOS
55	Li Dongyang (2021)	Single Source	CERT r4.2 & r6.2
56	Li Ximing (2023)	Single Source	CERT r4.2 & r6.2
57	Li Ximing (2024)	Multi-Source	CERT + UMDWikipedia
58	Lilhore et al. (2025)	Multi-Source	CIC-IoT 2023 + UNSW-NB15
59	Liu et al. (2025)	Multi-Source	CERT + DARPA1999
60	Manoharan (2024)	Single Source	CERT r4.2
61	Matos et al. (2025)	Single Source	Synthetic Healthcare Logs
62	Medvedev et al. (2025)	Multi-Source	CMU + KeyRecs
63	Mehmood et al. (2023)	Single Source	CERT (Customized)
64	Mehnaz (2021)	Single Source	Wikipedia File Repository Logs
65	Mladenovic (2024)	Single Source	CERT r4.2
66	Nasir et al. (2021)	Single Source	CERT r4.2
67	Nikiforova (2024)	Single Source	Real-world Audit Logs (Proprietary)
68	Pal et al. (2023)	Single Source	CERT r4.2, r5.2, r6.2
69	Patel (2025)	Single Source	Landauer UEBA Log Dataset
70	Peccatiello (2023)	Single Source	CERT v4.2
71	Pérez-Gomariz (2025)	Single Source	Fortune 500 Incident Logs
72	Prasad (2024) [Enhanced]	Single Source	CERT v5.2
73	Prasad (2025) [Hybrid]	Single Source	CERT
74	Prasad (2025) [Deep]	Single Source	CERT
75	Qawasmeh (2025)	Single Source	Synthetic Organizational Dataset
76	Qi et al. (2025)	Multi-Source	CERT r4.2 + LANL
77	Racherache (2023)	Single Source	Real-world Network Traffic (Industry Partners)
78	Randive (2023a)	Single Source	CERT r4.2
79	Randive (2023b)	Single Source	CERT r4.2 & r5.2
80	Rauf et al. (2021)	Single Source	CERT v4.2

81	Saminathan (2023)	Single Source	Simulated Organizational Environment
82	Bin Sarhan (2023)	Single Source	CERT r4.2
83	Senevirathna (2025)	Multi-Source	CERT + Video CCTV Data
84	Singh et al. (2022)	Single Source	CERT v4.2
85	Singh et al. (2023)	Single Source	CERT r4.2
86	Sivakrishna (2025)	Single Source	CERT r4.2, r5.2, r6.2
87	Song et al. (2024)	Single Source	CERT v6.2
88	Tabassum (2024)	Single Source	UK Hospital EHR Logs
89	Tao (2023)	Single Source	SapiMouse
90	Tao (2025)	Single Source	CERT v4.2
91	Tian Zhihong (2024)	Single Source	CERT v6.2
92	Tian Tian (2025)	Single Source	CERT v6.2
93	Villarreal-Vasquez (2023)	Single Source	Commercial EDR System Logs
94	Wall (2021)	Single Source	CERT r4.1
95	Wang Cheng and Zhu (2022)	Multi-Source	CERT + Banking + Kyoto + Darknet + Gowalla
96	Wang & El Saddik (2023)	Single Source	CERT r4.2 & r6.2
97	Wang Jiarong et al deep (2023)	Single Source	CERT r4.2
98	Wang et al Fusion(2022)	Single Source	CERT r4.2 (HTTP)
99	Wang (2024)	Single Source	CERT r6.2
100	Wei Yichen (2021)	Multi-Source	CERT r6.2 + KDD CUP 99
101	Wei Zhiyuan (2024)	Single Source	CERT r4.2
102	Wen et al. (2023)	Single Source	Enron Email
103	Xiao Junchao (2023)	Single Source	CERT r6.2
104	Xiao Haitao (2024)	Single Source	CERT
105	Xiao Fengrui (2025)	Multi-Source	LANL + CERT r6.2 + TWOS
106	Ye Xiaoyun (2025)	Single Source	CERT v4.2 & v5.2
107	Yi & Tian (2024)	Single Source	CERT r4.2
108	Yumlembam (2025)	Single Source	CERT r5.2 & r6.2
109	Zhang et al. (2021)	Single Source	CERT v4.2 & v6.2
110	Zhu et al. (2024)	Single Source	CERT

7. Dataset Source Grouping by Study Category

The following tables list the studies within each high-level dataset group. These group-level tables make it easier to locate studies that used CERT only, multi-source fusion, real-world/proprietary logs, or other specialized datasets:

Group 1: Single Source: CERT Benchmark (n=67)

The table below confirms the concentration of the reviewed literature around CERT benchmark versions, especially r4.2, r5.2, and r6.2.

#	Author (Year)	#	Author (Year)	#	Author (Year)
1	Adun (2023)	24	Gayathri R (2025)	47	Rauf et al. (2021)
2	Al-Mhiqani (2021)	25	Hong (2023)	48	Singh et al. (2022)
3	Al-Mhiqani (2022)	26	Jaiswal (2024)	49	Singh et al. (2023)
4	Almusawy (2024)	27	Jyosthna (2021)	50	Sivakrishna (2025)
5	Al-Shehari & Alsawail (2021)	28	Kabir (2023)	51	Song et al. (2024)
6	Al-shehri Abdullah (2022)	29	Kan (2023)	52	Tao (2025)
7	Al-Shehari (2023) [IF]	30	Kim (2023)	53	Tian Zhihong (2024)
8	Al-Shehari (2023) [Res]	31	Kotb (2025)	54	Tian Tian (2025)
9	Al-Shehari (2024) [CNN]	32	Le (2021) [Exploring]	55	Wall (2021)
10	Al-Shehari (2024) [LOF]	33	Li Dongyang (2021)	56	Wang & El Saddik (2023)
11	Alslaiman (2023)	34	Li Ximing (2023)	57	Wang Jiarong deep cluster (2023)
12	Amiri-Zarandi (2023)	35	Manoharan (2024)	58	Wang et al. Fusion (2022)
13	Amuda (2022)	36	Mehmood et al. (2023)	59	Wang (2024)
14	Anju (2024)	37	Mladenovic (2024)	60	Wei Zhiyuan (2024)
15	Asha (2024)	38	Nasir et al. (2021)	61	Xiao Junchao (2023)
16	Bin Sarhan (2023)	39	Pal et al. (2023)	62	Xiao Haitao (2024)
17	Cai (2024)	40	Peccatiello (2023)	63	Ye Xiaoyun (2025)
18	Ding (2024)	41	Prasad (2024) [Enhanced]	64	Yi & Tian (2024)
19	Dong (2025)	42	Prasad (2025) [Hybrid]	65	Yumlembam (2025)
20	Dosh (2021)	43	Prasad (2025) [Deep]	66	Zhang et al. (2021)
21	Duan (2024)	44	Rahman (2022)	67	Zhu et al. (2024)
22	Feng (2025)	45	Randive (2023a)		
23	Gayathri (2024)	46	Randive (2023b)		

Group 2: Multi-Source Fusion (n=23)

The next table shows how researchers combine CERT with LANL, TWOS, Enron, NSL-KDD, UNSW-NB15, IoT, biometric, and domain-specific datasets to broaden behavioral coverage.

#	Author (Year)	Specific Datasets Combined
1	Anakath (2022)	Open-source Biometric + Simulated Logs
2	Budžys (2024)	CMU Keystroke + GREYC-NISLAB
3	Fei (2023)	CERT r4.2 + LANL
4	Ferraro (2025)	PicoDomain + CERT r5.2
5	Gayathri B (2025)	CERT + NSL-KDD + UNSW-NB15
6	Gonzales (2025)	Synthetic Action Databases + UniProt
7	He (2022)	Enron + monkey.org + CERT r6.2

8	He (2024)	Enron + monkey.org + CERT r6.2
9	Huang (2025)	KDD-UEBA + NSL-UEBA + CIC-UEBA + KDD-UNBLogs
10	Janjua (2021)	Enron + TWOS
11	Kamatchi (2025)	Simulated (NS-3) + X-IIoTID
12	Kong (2025)	CERT r4.2 + SEA + HCT + PU
13	Lavanya (2024)	CERT r6.1 & r6.2 and IoT enabled institutional log records
14	Le (2021) [Ensembles]	CERT + LANL + TWOS
15	Li Ximing (2024)	CERT + UMDWikipedia
16	Lilhore et al. (2025)	CIC-IoT 2023 + UNSW-NB15
17	Liu et al. (2025)	CERT + DARPA1999
18	Medvedev et al. (2025)	CMU + KeyRecs
19	Qi et al. (2025)	CERT r4.2 + LANL
20	Senevirathna (2025)	CERT + Video CCTV
21	Wang Cheng (2022)	CERT + Banking + Kyoto + Darknet + Gowalla
22	Wei Yichen (2021)	CERT r6.2 + KDD CUP 99
23	Xiao Fengrui (2025)	LANL + CERT r6.2 + TWOS

Group 3: Single Source: Real-world / Proprietary (n=10)

This table identifies studies that used real-world or proprietary operational data. These studies provide stronger realism but are harder to replicate because the datasets are usually private, anonymized, or organization-specific.

#	Author (Year)	Source Type
1	Ahmad (2025)	Real-world Production Logs (Windows/Wazuh)
2	Ahmadi (2025)	Anonymized Corporate Logs
3	Ali (2025)	US Army Insider Threat Hub (Narrative)
4	Erola (2022)	Enterprise Logs (Proxy, VPN, AD from 3 companies)
5	Hurst (2022)	UK Hospital EHR Logs
6	Nikiforova (2024)	Proprietary Business Audit Logs
7	Pérez-Gomariz (2025)	Fortune 500 Incident Logs
8	Tabassum (2024)	UK Hospital EHR Logs
9	Villarreal-Vasquez (2023)	Commercial EDR System Telemetry
10	Racherache (2023)	Real-world Network Traffic

Group 4: Single Source: Other / Specialized Synthetic (n=10)

The studies in this table use specialized sources such as Enron, SOFIT, SapiMouse, synthetic organizational logs, healthcare simulations, and UEBA testbeds. These datasets are useful for specific research goals but may not represent the full complexity of enterprise insider threat behavior.

#	Author (Year)	Specialized Source
1	Alabdulkreem (2022)	Custom Aggregated Internal Logs
2	Greitzer (2021)	SOFIT Ontology Testbed

3	Haq (2022)	Enron Email Corpus
4	Matos et al. (2025)	Synthetic Healthcare Access Logs
5	Mehnaz (2021)	Wikipedia File Repository Logs
6	Patel (2025)	Landauer UEBA Log Dataset
7	Qawasmeh (2025)	Synthetic Organizational Dataset
8	Saminathan (2023)	Simulated Organizational Logs
9	Wen et al. (2023)	Enron Email Corpus
10	Tao (2023)	SapiMouse (Biometric)

Overall, the dataset distribution shows that ML-based insider threat detection research is strongly shaped by dataset availability. Public benchmarks such as CERT support reproducibility and comparison, but they may not fully represent real organizational behavior. Real-world and proprietary datasets improve realism but remain difficult to access and reproduce. Multi-source and specialized datasets expand detection coverage, but they introduce additional challenges related to heterogeneity, labeling, preprocessing, and cross-dataset generalization.